Dataset Reference:
https://drive.google.com/file/d/1cCB19GQ3Of92RAk5TVb0DPohCt1eZR6d/view?usp=sharing

Question 1: Exploratory Data Analysis (EDA) -20

Problem Statement:
Find the three most important factors that separate individuals who earn more than $50K annually from those who earn $50K or less.

Tasks:

- Perform EDA using appropriate charts and basic descriptive statistics.
- Select your top 3 features and justify your choices using visual and statistical evidence.
- For each selected feature, explain in simple terms why it intuitively affects income.
- Rank your 3 features from most important to least important, provide why you think that ?

Question 2: Preprocessing Pipeline Design (Regression) - 20

Problem Statement:
Predicting a continuous variable requires clean, properly formatted data. You will design a preprocessing pipeline to prepare the dataset for predicting **hours-per-week** based on the other available features.

Tasks:

- Design Scikit-Learn Pipelines that handles missing values, categorical feature encoding, and numerical feature scaling.
- Documentation: Clearly explain what specific preprocessing steps you applied to which columns and why you chose those specific methods.

Question 3: Regression Modeling & Evaluation - 20

Problem Statement:
Utilizing the preprocessing pipeline designed in Question 2, train two different regression models to predict **hours-per-week** and evaluate their performance.

Tasks:

- Train a standard Linear Regression model using your pipeline.
- Train a Stochastic Gradient Descent (SGD) Regressor model using your pipeline.
- Evaluate and compare both models using R2, MAE, and MSE.
- Briefly discuss which model performed better and why.

Question 4: Classification Pipeline & Error Analysis -20

Problem Statement:
Shift your focus from regression back to classification. Build an end-to-end Logistic Regression pipeline to predict whether a person earns more than $50K annually (income feature)

Tasks:

- Construct a new Scikit-Learn Pipeline that includes your new preprocessing steps and a Logistic Regression classifier.
- Train the model and evaluate its performance on a test set using Accuracy, Precision, and Recall.

Question 5: Hyperparameter Tuning & Model Comparison

Problem Statement:
Train KNN model to classify income column. Determine if a distance based model like K-Nearest Neighbors (KNN) can outperform your Logistic Regression model when properly optimized.

Tasks:

- Use GridSearchCV to systematically tune a KNN Classifier on hyperparameters: n_neighbors, weights, and metric.
- Once the optimal KNN model is found, evaluate it using Accuracy, Precision, and Recall.
- Comparison: Compare the performance metrics of your Tuned KNN model against the default Logistic Regression model. Which algorithm performed better overall.